

VR4Parenting: Using Virtual Reality to Support Positive Parenting and Prevent Violence Against Children

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Abstract

Violence against children remains a global public health concern, with parents and caregivers often identified as perpetrators. Although evidence-based parenting interventions exist, many rely heavily on theoretical instruction and provide limited opportunities for embodied, emotionally grounded practice. VR4Parenting is an ongoing research project exploring how immersive virtual reality (VR) can augment positive parenting interventions through experiential learning, embodied perspective-taking, and skills rehearsal within a safe and controlled environment. The project combines participatory co-design with parents and practitioners, the development of a 360° video-based VR prototype, and a planned multi-phase evaluation protocol. Rather than reporting empirical findings, the paper focuses on the design rationale and methodological choices underpinning the prototype and its evaluation, including usability, acceptability, experiential outcomes, as well as its potential to support reflection and perspective-taking in parenting contexts. By outlining a structured and ethically grounded research protocol, this work contributes early design insights and methodological guidance for the CHI community on designing and evaluating immersive systems in socially relevant contexts.

CCS Concepts

• Human-centered computing; • Virtual reality; Participatory design; • Applied computing; • Psychology;

Keywords

Virtual Reality, Positive Parenting, Embodied Learning, Perspective-Taking, Participatory Design

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1 Introduction

Violence against children remains a major global public health concern, with approximately one in two children aged 2–17 experiencing some form of violence each year, most often perpetrated by parents or caregivers [1–4]. Exposure to family violence is associated with significant emotional, behavioral, and relational difficulties and increases the risk of intergenerational transmission of violent practices [5–7].

In response, numerous parenting interventions have been developed to promote non-violent disciplinary strategies. Although meta-analyses indicate short-term effectiveness [8,4], long-term impacts remain limited [9]. Many programs rely heavily on psychoeducation and behavioral techniques that parents are expected to implement independently, offering limited opportunities to engage with the emotional and embodied processes that shape real-world parent–child interactions [10,11].

Immersive virtual reality (VR) offers a complementary approach by enabling embodied perspective-taking, allowing users to experience situations from another person’s first-person perspective with a strong sense of presence and agency [12,13]. Prior work demonstrates that embodied VR can influence empathy, social cognition, and behavior, including in violence-prevention contexts [14–18]. More recently, VR has been used to simulate the perceptual perspective of a child, with promising effects on parenting-related empathy [19–21]. However, such applications remain limited and are rarely embedded within structured parenting interventions. Moreover, most prior studies have focused primarily on mothers, despite the importance of co-parenting in family functioning.

In this paper, we present VR4Parenting, an ongoing research project exploring how immersive VR can augment parenting interventions by fostering empathy, self-regulation, and reflective awareness. We position this work within the CHI community to



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share early design insights and a study protocol, and to invite discussion on the ethical and experiential design of immersive systems for family wellbeing.

2 Method

2.1 Study Design

This work will follow a multi-phase, mixed-methods design that combines participatory design, immersive system development, and pilot evaluation. The project aims to co-design, develop, and conduct a preliminary evaluation of the immersive parenting simulations created as part of the VR4Parenting project. The approach will draw on established HCI methods, including stakeholder co-design, usability testing, and exploratory evaluation of experiential outcomes, alongside well-validated psychological assessments to examine the potential benefits of incorporating these types of experiences into existing parenting interventions and therapeutic contexts.

Within this framework, the mixed-methods and participatory strategy is explicitly grounded in evidence-based parenting interventions and psychological theories of parenting and children wellbeing, ensuring a strong top-down, theory-informed foundation for the development of the VR parenting scenarios. However, because parenting practices unfold within emotionally charged, context-dependent family interactions, often shaped by stress, child behavior, cultural expectations, and daily routines, relying solely on theory-driven design may overlook important real-life nuances of parent-child dynamics. For this reason, we plan to complement the theoretical framework with direct input from parents and practitioners to ensure that the simulated scenarios reflect authentic parenting challenges and responses, thereby enhancing ecological validity, experiential realism, and overall acceptability of the designed VR scenarios.

More specifically, VR4Parenting project is comprised by three main phases based on: (1) participatory co-design with parents and professionals (e.g., psychologists, educators, health professionals, research experts in related topics) to inform the development of ecologically valid VR scenarios; (2) development of a VR prototype and a laboratory-based pilot study with parents experiencing the immersive parenting scenarios; and (3) an evaluation with practitioners to assess usability, acceptability, and applicability of these immersive experiences in real-world contexts (Figure 1).



Figure 1: Overview of the three study phases of VR4Parenting project.

2.2 Participants

Inclusion criteria for parent participants will include being between 18 and 60 years old, being a parent of a child within the target age range defined during Phase 1. Exclusion criteria will include epilepsy, severe visual or motor impairments, and prior convictions for violent crimes. Professional participants will be recruited based on their expertise in parenting, education, or family intervention contexts. All participants will receive detailed information about the study and will participate only after providing informed consent.

2.3 Measures

In Phase 1, data will be collected through an online self-report survey and in-person focus groups. The survey will capture challenging parenting situations, strategies used to deal with these situations, and basic sociodemographic information to guide scenario selection. Focus groups will use participatory design activities and semi-structured interviews to explore scenario narratives, interaction dynamics, and emotional relevance. Focus groups sessions will be audio-recorded with consent and analyzed thematically to extract design requirements.

In Phase 2, we will conduct a pilot study with parents using a mixed-methods approach. Quantitative data will be collected through standardized self-report questionnaires administered before and after the VR experience to assess key experiential mechanisms and parenting-related outcomes. Embodiment and presence will be measured through validated questionnaires to examine whether participants experience a sense of ownership over the child's or the simulated parent's perspective and psychological immersion within the scenario. These constructs are central to our learning hypothesis, as embodied perspective-taking is expected to facilitate emotional insight and empathic understanding.

Empathy-related changes will be assessed using instruments such as the Reading the Mind in the Eyes Test, which may capture cognitive components of perspective-taking, alongside affective self-report measures. Emotional arousal and valence during VR exposure will be assessed using measures such as the Self-Assessment Manikin (SAM). Parenting-related constructs, including parental self-efficacy and perceived competence, will be measured using instruments such as the Parenting Sense of Competence Scale. Changes in perceived child behavior and parental stress will be explored using measures such as the Eyberg Child Behavior Inventory, providing preliminary insight into whether experiential shifts translate into parenting-relevant cognitions.

Finally, usability will be assessed to ensure that any observed psychological effects are not confounded by interface barriers. Ad hoc questionnaires will capture perceived usefulness, emotional impact, and reflective insight immediately following the experience. Behavioral observations during the VR experience will also be recorded.

Together, these measures will enable us to examine whether the VR system achieves its intended experiential goals, namely embodied perspective-taking, emotional engagement, and reflective awareness, and whether these mechanisms demonstrate preliminary associations with parenting-related outcomes.

In Phase 3, practitioners will complete structured questionnaires assessing usability, acceptability, and perceived applicability of the

VR system, followed by brief interviews exploring feasibility, integration into existing programs, and opportunities for refinement. Assessment of satisfaction, perceived usability, and relevance of the VR prototype, will be measured with validated scales such as System Usability Scale and USE Questionnaire. Interviews will be audio-recorded with informed consent and analyzed qualitatively.

All data will be completely anonymized, and participants will be able to stop their participation in the studies at any time, without the need to provide explanations.

2.4 System Implementation

In our first prototype development of the immersive VR4Parenting scenes we plan to use 360° video visualized on an Oculus Quest 3 headset or similar. The VR experience will present users with realistic first-person scenarios depicting everyday parent–child interactions. Professional actors will be hired to perform the scenes.

The VR scenes will focus on common and emotionally salient parenting challenges that reflect everyday parent–child interactions. Although interaction is inherently limited in 360° video compared to fully interactive virtual environments, we plan to implement design strategies to mitigate these constraints and enhance perceived agency. First, dialogues will be explicitly directed at the participant, with characters addressing them by role, thereby clearly establishing their embodied position within the scene. Narrative branching will also be incorporated at key moments, allowing participants to select between predefined response options that shape the progression of the interaction, thus supporting decision-making and reflective engagement despite the non-fully interactive format. Additionally, at the end of the experience, guided reflection will be included to encourage active cognitive and emotional participation rather than only passive observation.

To strengthen embodiment, we will build on prior research demonstrating that full-body ownership illusions can be elicited using 360° video through first-person camera perspectives [22] (Figure 2). Finally, scene content and pacing will be carefully designed to minimize discomfort.

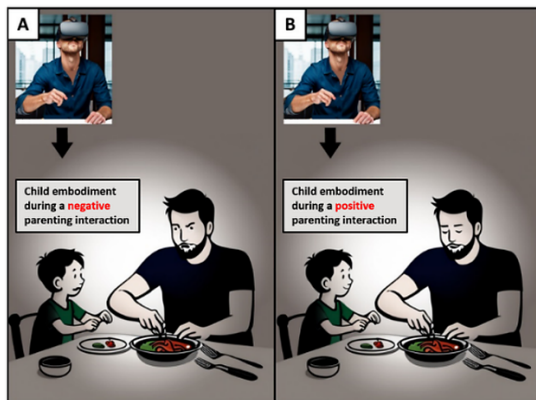


Figure 2: Conceptual illustration of a VR simulation in which a parent experiences a father-son interaction from the child’s perspective (positive or negative) and practices parenting skills from the parent’s perspective. The specific scenarios to be simulated will be determined through the participatory design process.

3 Expected Results and Future Research Directions

As this paper presents a study protocol for a first prototype of the VR4Parenting application, empirical results are still not available at the time of submission. The planned analyses will focus on assessing the usability, acceptability, and experiential impact of the VR4Parenting prototype, as well as its potential to support reflection and perspective-taking in parenting contexts.

We expect parents to report high levels of presence and perspective-taking during the VR experience, indicating that the scenarios are perceived as immersive and ecologically valid. Experiencing parenting situations from a simulated child’s perspective is anticipated to foster greater empathic understanding and increased awareness of children’s emotional states, compared to experiencing the same situations solely from the parent’s perspective. Additionally, practicing specific skills from a parent’s point of view within an immersive scenario is expected to enhance learning outcomes relative to theoretical explanations alone. Finally, we anticipate positive usability and perceived usefulness ratings, suggesting that the VR system is both feasible and acceptable for use in parenting support and training contexts.

Qualitative analyses of post-experience interviews are expected to reveal rich insights into participants’ emotional engagement, perceived realism, and reflections on their own parenting behaviors. These data are anticipated to highlight design elements that effectively support learning and self-reflection, as well as aspects of the VR experience that may require refinement to ensure emotional safety and usability.

From the practitioner perspective, we expect feedback to indicate high perceived relevance of the VR tool for complementing existing parenting interventions, along with practical considerations regarding implementation, session structure, and integration into professional practice. Together, these outcomes will inform iterative improvements of the VR prototype and guide the design of future large-scale and longitudinal evaluations.

If the findings of this pilot study are promising, our long-term objective will be to conduct larger-scale, controlled trials involving more diverse samples of parents. These studies will incorporate comparison groups to systematically evaluate whether parenting interventions that integrate immersive VR simulations yield greater benefits than traditional, non-immersive approaches, on the short and long-term. Such research will allow us to determine whether experiential VR components meaningfully enhance the impact of positive parenting programs. Effectiveness will be assessed through multiple indicators, including parents’ perceived self-efficacy and confidence in managing challenging parent–child situations, increased use of positive and non-coercive parenting strategies, reduced reliance on harsh or violent disciplinary practices, and improvements in both parental and child well-being. Through this extended research agenda, we aim to rigorously examine whether immersive simulations can sustainably and measurably strengthen the effectiveness of existing parenting interventions.

In this initial pilot phase, children will not be included in the participatory design process due to the methodological complexity and significant ethical considerations involved in engaging minors in research on sensitive parenting and violence-related situations. Given

the need to ensure emotional safety and appropriate safeguards, the current study will focus exclusively on adult participants. However, if the pilot outcomes are encouraging, future phases of the project will aim to incorporate children's perspectives in developmentally appropriate and ethically sound ways to co-develop new scenarios. Including children in subsequent participatory stages will help us enhance the relational authenticity, developmental sensitivity, and ecological validity of the immersive parenting simulations.

4 Conclusions

This paper presents the design rationale and planned evaluation protocol for VR4Parenting, a virtual reality–based prototype aimed at supporting positive parenting practices through embodied perspective-taking. By combining participatory design methods with immersive VR technology, the work outlines how emotionally sensitive parent–child scenarios can be represented in an ethically grounded, user-centered manner, with particular attention to realism, the potential to fit within current parenting interventions, and acceptability.

The proposed study contributes a structured protocol for the co-design, development, and preliminary assessment of VR experiences targeting parenting and family wellbeing. Rather than reporting empirical outcomes, the paper focuses on the methodological decisions underlying the prototype and evaluation design, offering guidance on how immersive systems can be systematically explored in contexts involving complex interpersonal and emotional dynamics.

By detailing a rigorous and ethically informed research plan, this work lays the groundwork for future empirical studies examining the feasibility, experiential qualities, and potential learning benefits of immersive perspective-taking in parenting contexts. More broadly, it aims to inform the CHI community on the design and evaluation of human-centered VR interventions for social good and their integration into existing support and training frameworks.

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